

Yuanyuan Niu

Gender: Male **Nationality:** Chinese

Date of Birth: 17 December 1996

Address:

Singapore: Department of the Built Environment, National
University of Singapore, 2 Architecture Drive,
Singapore, 117566

China: Southeast University, 2 Sipailou, Nanjing City,
Jiangsu Province, China, 210096

Telephone: (+65)93769304 / (+86)18805167230

E-mail: yuanyuanniu1217@gmail.com / yuanyuanniu@u.nus.edu



Education

-
- | | |
|---|--|
| ✧ School of Energy and Environment, Southeast University | (Nanjing, China) |
| <i>Ph.D. Candidate IN Power Engineering and Engineering Thermophysics</i> | <i>Sep. 2022 – Present</i> |
| | <i>Supervisor: Prof. Hua Qian</i> |
| ✧ College of Design and Engineering, National University of Singapore | (Singapore) |
| <i>Visiting Ph.D. Candidate IN Built Environment</i> | <i>Jan. 2026 – Present</i> |
| | <i>Supervisor: Prof. Ali Ghahramani & Prof. Cheong Kok Wai David</i> |
| ✧ School of Energy and Power Engineering, Nanjing University of Science and Technology(Nanjing, China) | |
| <i>M.Sc. IN Heating, Gas Supply, Ventilating and Air Conditioning Engineering</i> | <i>Sep. 2019 – Jun. 2022</i> |
| | <i>Supervisor: Prof. Fang Wang</i> |
| ✧ School of Energy and Power Engineering, Nanjing University of Science and Technology(Nanjing, China) | |
| <i>B.E. IN Built Environment and Energy Application Engineering</i> | <i>Sep. 2015 – Jun. 2019</i> |
| | <i>Supervisor: Prof. Fang Wang</i> |

Publications

-
- [1] **Yuanyuan Niu**, Xiaohong Zheng, Jiayu Huang, Chenxue Song, Danting Luo, Haoren Qiu, Yongsheng Zhou, Hua Qian, Spread of bidet toilet flushing-generated infectious aerosols in the cubicle of a public restroom, *Building and Environment* 275 (2025) 112844. **(IF=7.2; Q1)**.
- [2] **Yuanyuan Niu**, Fang Wang, Danting Luo, Zhiyong Shu, Jiayu Huang, Yongpeng Zhang, Cong Liu, Hua Qian, Vertical transmission of infectious aerosols through building toilet drainage system: An experimental study, *Environmental Pollution* 343 (2024) 123284. **(IF=7.6; Q1)**
- [3] **Yuanyuan Niu**, Fang Wang, Songmin Liu, Wenjie Zhang, Source analysis of heavy metal elements of PM2.5 in canteen in a university in winter, *Atmospheric Environment* 244 (2021) 117879. **(IF=4.2; Q1)**
- [4] **Yuanyuan Niu**, Hua Qian, Xiaohong Zheng, Jiayu Huang, Chenxue Song, Toilet lids: Protectors preventing the spread of infectious bioaerosols generated during flushing bidet toilet, *Indoor Air*, Honolulu, U.S.A, 2024.
- [5] **Yuanyuan Niu**, Fang Wang, Danting Luo, Zhiyong Shu, Jiayu Huang, Yongpeng Zhang, Cong Liu, Hua Qian, Simulated experimental study on SARS-CoV-2 transmission in building toilet drainage system, *Healthy Buildings*, Tianjin, China, 2023.
- [6] Songmin Liu, Fang Wang, **Yuanyuan Niu**, Wenjie Zhang, Influence of window angle at night on indoor air quality and sleep quality, *Building Energy & Environment* 41(5) (2022) 28-33. **(In Chinese)**

- [7] **Yuanyuan Niu**, Analysis of aircraft air conditioning system status and development direction, Shanxi Architecture 44(16) (2018) 95-96. **(In Chinese)**
- [8] **Yuanyuan Niu**, An analysis of the relations between the Siheyuan form in Beijing and the environment in Beijing, Shanxi Architecture 44(15) (2018) 22-23. **(In Chinese)**
- [9] **Yuanyuan Niu**, Research method and route of building energy saving technology in expressway service area, Shanxi Architecture 44(12) (2018) 191-192. **(In Chinese)**
- [10] **Yuanyuan Niu**, Research significance and present situation of building energy saving technology in expressway service area, Shanxi Architecture 44(11) (2018) 216-217. **(In Chinese)**

Honors and Awards

- | | |
|--|--|
| 1. National Scholarship , Ministry of Education of the People's Republic of China | <i>Dec.2021</i> |
| 2. National Scholarship , Ministry of Education of the People's Republic of China | <i>Nov.2018</i> |
| 3. China Scholarship Council – PhD Fellowship for High-Level University Postgraduate Students , China Scholarship Council | <i>May 2025</i> |
| 4. The Second Prize of the China Refrigeration and Air Conditioning Industry College Student Science and Technology Competition , China Refrigeration and Air-Conditioning Industry Association | <i>Jul.2018</i> |
| 5. The Third Prize in the Youth Member Innovation and Entrepreneurship Competition of the Jiangsu Refrigeration Society , Jiangsu Association of Refrigeration | <i>Oct.2018</i> |
| 6. The Third Prize in the "Internet Plus" College Student Innovation and Entrepreneurship Competition , Nanjing University of Science and Technology | <i>Sep.2020</i> |
| 7. The Third Prize in the "Challenge Cup" Science and Technology Competition , Nanjing University of Science and Technology | <i>Sep.2020</i> |
| 8. The Second Prize Scholarship , Southeast University | <i>Nov.2025</i> |
| 9. The Second Prize Scholarship , Southeast University | <i>Nov.2024</i> |
| 10. The Third Prize Scholarship , Southeast University | <i>Oct.2023</i> |
| 11. The First Prize Scholarship , Southeast University | <i>Oct.2022</i> |
| 12. The First Prize Scholarship , Nanjing University of Science and Technology | <i>Nov.2021</i> |
| 13. The First Prize Scholarship , Nanjing University of Science and Technology | <i>Nov.2020</i> |
| 14. The First Prize Scholarship , Nanjing University of Science and Technology | <i>Oct.2019</i> |
| 15. The Second Prize Scholarship , Nanjing University of Science and Technology | <i>Oct.2018</i> |
| 16. The Second Prize Scholarship , Nanjing University of Science and Technology | <i>Oct.2017</i> |
| 17. The Second Prize Scholarship , Nanjing University of Science and Technology | <i>Mar.2017</i> |
| 18. The Third Prize Scholarship , Nanjing University of Science and Technology | <i>Oct.2016</i> |
| 19. The Third Prize Scholarship , Nanjing University of Science and Technology | <i>Mar.2016</i> |
| 20. Outstanding Graduate Student , Nanjing University of Science and Technology | <i>Nov.2021</i> |
| 21. Outstanding Graduate Student , Nanjing University of Science and Technology | <i>Nov.2020</i> |
| 22. Three Good Student , Nanjing University of Science and Technology | <i>Oct.2017</i> |
| 23. Completion Certificate of the Pioneer Program , Nanjing University of Science and Technology | <i>Nov.2017</i> |
| 24. Certificate of Reviewing , Building and Environment | <i>for 5 reviews between Jun.2021 and Dec.2023</i> |

Patent

✧ Hua Qian, **Yuanyuan Niu**, Xiaohong Zheng, Yubin Zhang, Jun Jiang, Yun Liu. A suction-blowing integrated toilet, December 16, 2025. China, CN121138413A (Patent Publication).

Conferences

1. **Indoor Air 2024**, Honolulu, Hawaii, U.S.A (Oral presentation) *Jul.2024*
2. **Healthy Buildings 2023 Asia**, Tianjin, China (Poster presentation) *Jul.2023*
3. **The 15th Nanjing Conference on International Exchange and Cooperation for Overseas Chinese Professionals**, Nanjing, China *Jun.2023*
4. **The 16th East China Academic Conference on Laboratory Animal Science**, Suzhou, China *Jun.2023*
5. **The 8th Member Representative Conference of the Jiangsu Refrigeration Society**, Nanjing, China *Oct.2018*

Academic Activities

✧ **Journal Reviewer:** Building and Environment (SCI).

✧ **Professional Society Memberships:** International Society of Indoor Air Quality and Climate (ISIAQ), Jiangsu Association of Refrigeration.

Skills

1. **Language:** English(CET-6: 453), Chinese (native).
2. **Programming:** Be skillful in Python and C++.
3. **Software:** Be proficient in Office, SPSS, Solidworks, AutoCAD, CFD, Style Builder, PhotoShop, Origin, Flowmaster, Quickshow, VOSviewer.